DS&A EXPERIMENT - 1

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PROGRAM :-

```
#include <stdio.h>
#include <string.h>

int length(char[]);
void compare(char[], char[]);
void palindrom(char[]);
void substring(char[], char[]);
void copy(char[], char[]);
void reverse(char[]);
void displayMenu();

int i, j;

int main()
{
    int choice;
    char str[50];
    char str2[50];
    int exit = 0;

    displayMenu();

    do
    {
        printf("\nENTER YOUR CHOICE: ");
        scanf("%d", &choice);
        getchar(); // Remove newline after scanf
```

```
if (choice >= 1 && choice <= 6)
{
    printf("Enter String: ");
    fgets(str, 50, stdin);
    str[strcspn(str, "\n")] = '\0';
}

switch (choice)
{
    case 1:
        printf("Length of the String is %d\n\n", length(str));
        break;

    case 2:
        printf("Enter second String: ");
        fgets(str2, 50, stdin);
        str2[strcspn(str2, "\n")] = '\0';

        copy(str, str2);
        break;

    case 3:
        reverse(str);
        break;

    case 4:
        palindrom(str);
        break;

    case 5:
        printf("Enter Substring: ");
        fgets(str2, 50, stdin);
        str2[strcspn(str2, "\n")] = '\0';
```

```
            substring(str, str2);
            break;

        case 6:
            printf("Enter second String: ");
            fgets(str2, 50, stdin);
            str2[strcspn(str2, "\n")] = '\0';

            compare(str, str2);
            break;

        case 7:
            displayMenu();
            break;

        case 8:
            exit = 1;
            printf("Program Ended.\n");
            break;

        default:
            printf("Invalid Choice!\n");
      }

   } while (exit == 0);

   return 0;
}

void displayMenu()
{
   printf("\n---------- MENU ----------\n");
   printf("1. Length\n");
   printf("2. Copy String\n");
   printf("3. Reverse String\n");
```

```
    printf("4. Check Palindrome\n");
    printf("5. Check Substring\n");
    printf("6. Compare Strings\n");
    printf("7. Display Menu\n");
    printf("8. Exit\n");
}

int length(char a[])
{
    int len = 0;

    while (a[len] != '\0')
    {
        len++;
    }

    return len;
}

void compare(char str[], char str2[])
{
    i = 0;

    while (str[i] == str2[i])
    {
        if (str[i] == '\0')
        {
            printf("Both Strings are Equal\n\n");
            return;
        }

        i++;
    }

    printf("Both Strings are Not Equal\n\n");
```

```
}

void palindrom(char a[])
{
   int l = length(a);
   int flag = 0;

   for (i = 0, j = l - 1; i < j; i++, j--)
   {
      if (a[i] != a[j])
      {
         flag = 1;
         break;
      }
   }

   if (flag == 0)
      printf("String is Palindrome\n\n");
   else
      printf("String is Not Palindrome\n\n");
}

void substring(char a[], char b[])
{
   int al = length(a);
   int bl = length(b);
   int flag = 0;

   if (bl > al)
   {
      printf("The String B is not a substring of A\n\n");
      return;
   }

   for (i = 0; i <= al - bl; i++)
```

```
    {
        for (j = 0; j < bl; j++)
        {
            if (a[i + j] != b[j])
                break;
        }

        if (j == bl)
            flag++;
    }

    if (flag > 0)
        printf("The String B is a substring of A and occurs %d time(s)\n\n",
flag);
    else
        printf("The String B is not a substring of A\n\n");
}

void copy(char a[], char b[])
{
    int bl = length(b);

    for (i = 0; i <= bl; i++)
    {
        a[i] = b[i];
    }

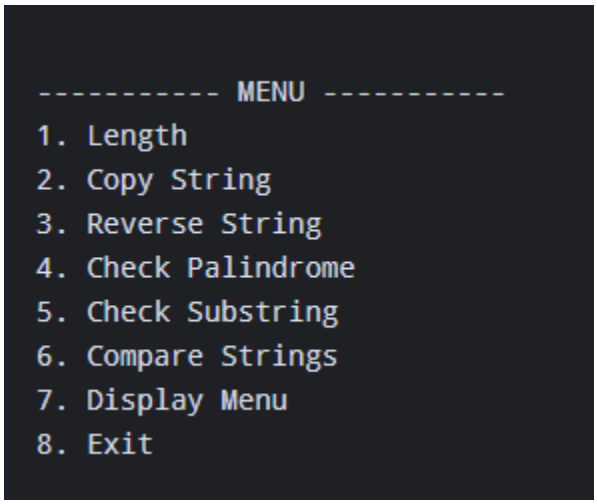
    printf("Copied String: %s\n\n", a);
}

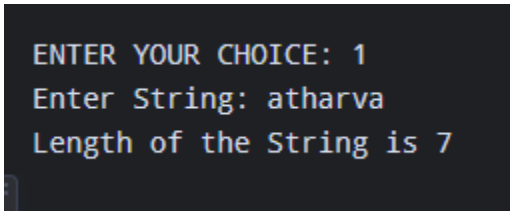
void reverse(char a[])
{
    int al = length(a);
    char temp;
```

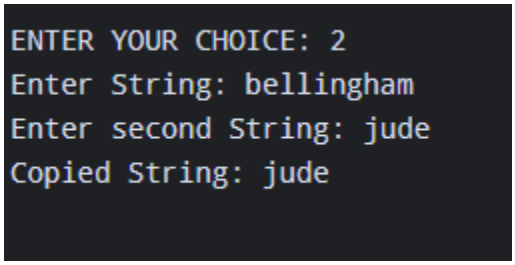
```
    for (i = 0, j = al - 1; i < j; i++, j--)
    {
        temp = a[i];
        a[i] = a[j];
        a[j] = temp;
    }

    printf("Reversed String: %s\n\n", a);
}
```

OUTPUT : -







```
ENTER YOUR CHOICE: 3
Enter String: rathod
Reversed String: dohtar
```

```
ENTER YOUR CHOICE: 4
Enter String: technology
String is Not Palindrome
```

```
ENTER YOUR CHOICE: 5
Enter String: addition
Enter Substring: add
The String B is a substring of A and occurs 1 time(s)
```

```
 ENTER YOUR CHOICE: 6
 Enter String: hello
 Enter second String: world
 Both Strings are Not Equal
```

```
ENTER YOUR CHOICE: 7

---------- MENU ----------
1. Length
2. Copy String
3. Reverse String
4. Check Palindrome
5. Check Substring
6. Compare Strings
7. Display Menu
8. Exit

ENTER YOUR CHOICE: 8
Program Ended.


=== Code Execution Successful ===
```